

Blockchain Applications

Lecture 4: Stablecoins I – Measuring Depeg Risk and On-Chain Early Warning



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
Master Blockchain for Business (MBB), FABIZ
24 November 2026

Today's Question and Route

- **Question**: a stablecoin promises one dollar. How far and how often does it miss, and can we see a miss coming?
  - stablecoins settle most crypto trades and serve as collateral in decentralised finance (DeFi): a deviation of a few basis points can trigger liquidations
- **Route**
  - measuring the peg: deviations in basis points, tails, episodes; the Silicon Valley Bank (SVB) weekend of March 2023
  - where the price is formed: automated market maker (AMM) invariants: constant product and StableSwap
  - early warning: event-level evaluation, utility, results of our paper
- Data: EODHD daily (2021–2026) and hourly (2022–2026) prices of USDT, USDC, DAI; DefiLlama supply (21 October 2026)

Learning Outcomes

- Measure a peg deviation in basis points and describe its distribution, tails and episodes
- Explain price impact under constant-product and StableSwap AMMs
- Explain why pool composition reveals liquidity stress before the price
- Evaluate an early-warning system with event-level recall, lead time, false alerts and utility

Abbreviations (1/2)

AMM automated market maker
AR autoregressive (model); AR(1): first order
ARIMA autoregressive integrated moving average
AUC area under the receiver operating characteristic curve
AUPRC area under the precision–recall curve
bn billion
bp basis point (0.01%)
BTC Bitcoin
CDF cumulative distribution function
CI confidence interval
CRPS continuous ranked probability score
DAI Maker/Sky stablecoin (crypto-backed)
DeFi decentralised finance
EODHD EOD Historical Data (data vendor)
ES expected shortfall
ETH Ether (Ethereum)
FABIZ Faculty of Business Administration in Foreign Languages, ASE
FDIC Federal Deposit Insurance Corporation
Fed Federal Reserve
FPCA functional principal component analysis
FRM Financial Risk Meter
GARCH generalised autoregressive conditional heteroskedasticity
GARCH-t GARCH with Student-t errors
GPD generalised Pareto distribution
HC heteroskedasticity-consistent
HHI Herfindahl–Hirschman index

Abbreviations (2/2)

JIT just-in-time (liquidity)
KS Kolmogorov–Smirnov (test)
KYC know your customer
m million
OLS ordinary least squares
PIT probability integral transform
PYUSD PayPal USD
QL quantile (pinball) loss
sd standard deviation
SE standard error
SHAP SHapley Additive exPlanations
SSR sum of squared residuals
SVB Silicon Valley Bank
TAR threshold autoregression
TiDE Time-series Dense Encoder
twCRPS threshold-weighted CRPS
USD US dollar
USD1 World Liberty Financial USD
USDC USD Coin (Circle)
USDe Ethena USDe
USDG Global Dollar (Paxos)
USDS Sky Dollar
USDT Tether
UTC Coordinated Universal Time
V3 version 3 (Uniswap V3)
VaR value-at-risk

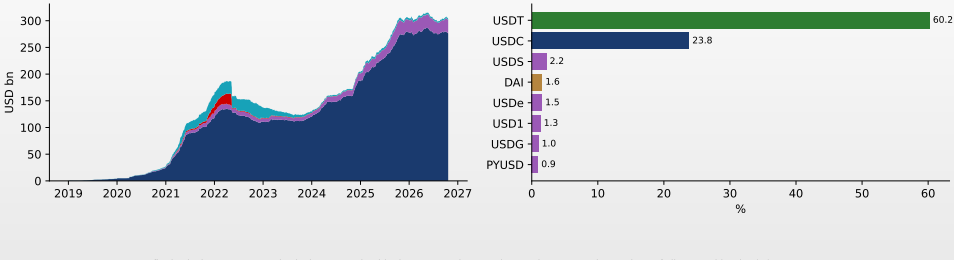
Reading and Tools

- Main paper: Chaffard, Găman, Staures, Ginavar, Jheng, Härdle & Pele (2026), *Onchain Monitoring of Stablecoin Depeg Risk*, preprint; data on Zenodo
- AMMs: Egorov (2019) (StableSwap), Adams et al. (2021) (Uniswap V3)
- Threshold models: Hansen (1996); forecast evaluation: Gneiting & Raftery (2007), Gneiting & Ranjan (2011); attribution: Lundberg & Lee (2017)
- Estimation: Python (`numpy`, `scipy`, `scikit-learn`); all numbers marked "own" are estimated for this lecture
- Next lecture (Stablecoins II): the De-pegging Pressure Index and the Financial Risk Meter (FRM), across many stablecoins

Mechanisms

- A stablecoin is a token that targets a fixed price P^* in a reference asset, here $P^* = 1$ USD
- **Fiat-backed** (USDT, USDC): reserves in cash and Treasury bills; the issuer mints and redeems at par for eligible clients
- **Crypto-backed** (DAI/USDS): over-collateralised loans in a smart contract; liquidations protect the peg
- **Algorithmic** (TerraUSD, May 2022): a mint/burn swap against a volatile token; no external collateral
- The peg is held by **arbitrage**: if $P_t < 1$, buy on the market and redeem at 1; if $P_t > 1$, mint at 1 and sell
  - arbitrage needs access (know-your-customer (KYC) rules, minimum size), time (bank hours) and trust in the reserves
  - when one of the three fails, the market price can leave the peg

Who Issues the Dollars on Chain



- Supply of USD stablecoins: 4.2 bn USD (Jan 2020), 187.1 bn (May 2022), 129.8 bn (Jan 2024), 303.4 bn (21 Oct 2026)
- Fiat-backed: 90.9%, crypto-backed 8.1%; algorithmic designs vanished after TerraUSD
- Two issuers hold 84% (USDT 60.2%, USDC 23.8%): a USDC/USDT dislocation is a system-level event

Three Distinctions

- **Fundamental vs market-defined peg**
  - fundamental: the redemption value; market-defined: the price on a given venue
  - a token has no single price: on 10 October 2025 USDe traded at 0.65 USD on one exchange while other venues stayed near 1
- **Exogenous vs endogenous**
  - exogenous: reserve losses, regulation, sanctions; arrive as news, hard to predict
  - endogenous: liquidity and order flow inside the market; visible on chain *before* the price moves
- **Transient vs persistent**
  - transient: hours, closed by arbitrage; persistent: a loss of confidence (TerraUSD)
  - a transient depeg is not harmless: liquidations at a bad price are irreversible
- This lecture: market-defined, mostly endogenous, often transient risk, where monitoring is feasible

Definitions

- Deviation in basis points (1 bp = 0.01%):

$$d_t = 10^4 \, (P_t - 1), \qquad d_t^{\mathrm{USDC/USDT}} = 10^4 \left(\frac{P_t^{\mathrm{USDC}}}{P_t^{\mathrm{USDT}}} - 1 \right)$$

  - closing price vs intraday low: the low shows the worst executable price, the close what remained
- Exceedance probability and breach indicator, threshold δ (here 15 bp):

$$\hat{P}(|d| \geq x) = \frac{1}{T} \sum_{t=1}^{T} \mathbb{1}\{|d_t| \geq x\}, \qquad b_t = \mathbb{1}\{|d_t| \geq \delta\}$$

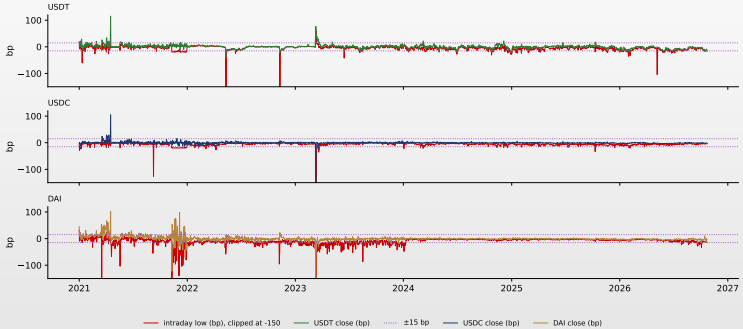
- An **episode** starts at the first breach hour after at least 24 hours without a breach
  - hours inside an episode are not independent observations: count episodes, not hours

Peg Deviations, Daily, 2021–2026 (own)

bp	mean	sd	$q_{0.01}$	$q_{0.99}$	min (date)	% $\lvert d\rvert \geq 15$	% $\lvert d\rvert \geq 50$	min low	kurt	$\hat{\rho}_1$
USDT	0.5	7.3	−15.5	21.3	−41 (11 May 22)	3.6	0.14	−515	40	0.73
USDC	−0.2	7.4	−7.6	9.7	−285 (11 Mar 23)	0.7	0.14	−1226	1045	0.29
DAI	0.4	10.8	−17.2	36.6	−261 (11 Mar 23)	5.2	0.57	−1030	185	0.34

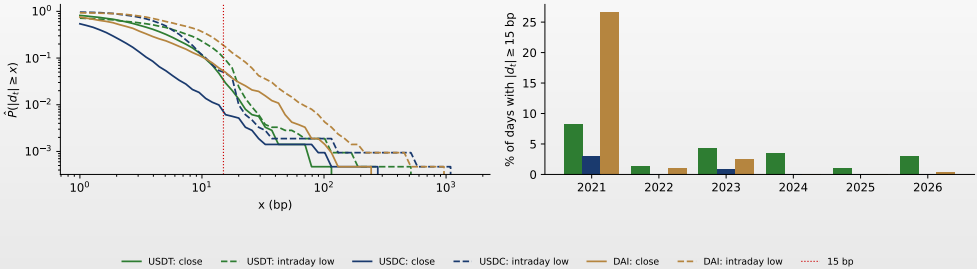
- 2 122 daily closes, EODHD aggregated prices; earlier years are excluded because the aggregate includes illiquid venues
- The mean is zero to the first decimal: the peg holds *on average*; the risk is in the tails
- USDC: kurtosis 1 045, one weekend dominates the distribution; its 1% quantile is only −7.6 bp
- USDT: smallest extreme, but most persistent ($\hat{\rho}_1 = 0.73$) and most often beyond 15 bp after 2022
- The worst intraday lows are 4 to 12 times deeper than the worst closes: a liquidation engine sees the low

The Three Dollars, Day by Day



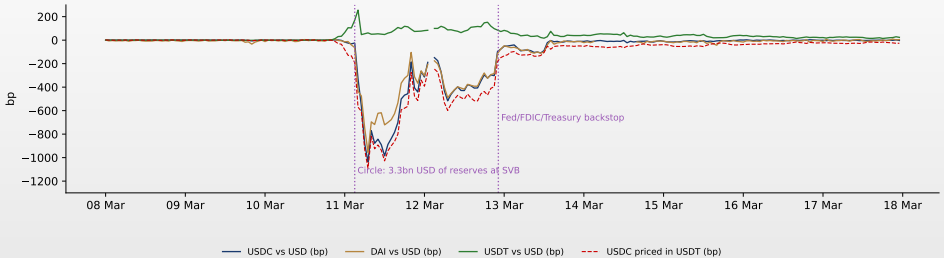
- Closes stay inside ±15 bp most of the time; the lows (red) reveal the stress days: May 2022 (Terra), March 2023 (SVB)

Tails and Their Frequency



- Left: on a log-log scale the exceedance curves decay roughly linearly beyond 15 bp, a power-law tail as in Lecture 2
- Right: DAI was beyond 15 bp on 26.6% of days in 2021, 0.3% in 2026; USDT is the only one still breaching regularly (3.0% in 2026)

The SVB Weekend, Hour by Hour (own)



- Hourly minimum: USDC −1 036 bp and DAI −955 bp, both at 07:00 Coordinated Universal Time (UTC) on Saturday 11 March 2023; USDT rose above 1 (flight to the other dollar)
- USDC back above −100 bp at 22:00 UTC on 12 March, just before the backstop; above −15 bp only on 13 March, 14:00 UTC
- DAI followed USDC because a large part of its collateral was USDC: contagion through the balance sheet

Worked Example: Was the SVB Discount Fair?

- Circle disclosed 3.3 bn USD of about 40 bn USD reserves at SVB: share $s = 3.3/40 = 8.25\%$
- With recovery rate R on the SVB deposit, the redemption value per token is

$$V(R) = 1 - s\,(1 - R), \qquad V(0) = 0.9175 \;\Rightarrow\; d = -825 \text{ bp}$$

- The market reached $-1\,036$ bp, below the value under *zero* recovery
- Interpretation
  - the extra 211 bp is a liquidity premium: banks were closed over the weekend, so redemption was impossible and arbitrage could not work
  - the discount was the price of time and uncertainty, not only of expected loss
  - a peg can fail with reserves that cover 92% or more: access matters as much as backing

Constant-Product AMM

- A pool holds x of token X and y of token Y; trades keep the invariant $x\,y = k$
- Marginal price of X in Y: $p = y/x$. Selling Δx returns

$$\Delta y = y - \frac{k}{x + \Delta x} = \frac{y\,\Delta x}{x + \Delta x}, \qquad \text{slippage} = 1 - \frac{\Delta y}{\Delta x} = \frac{\Delta x}{x + \Delta x} \quad (p = 1)$$

- Worked example: pool 10 m X + 10 m Y, sell 1 m X

$$\Delta y = \frac{10 \cdot 1}{11} = 0.909 \text{ m}, \qquad \text{slippage } 9.1\%$$

- Liquidity is spread over all prices from 0 to ∞: robust but very expensive for stable pairs

StableSwap (Curve): Flat Near Parity

- Invariant for n coins, amplification A (Egorov (2019)):

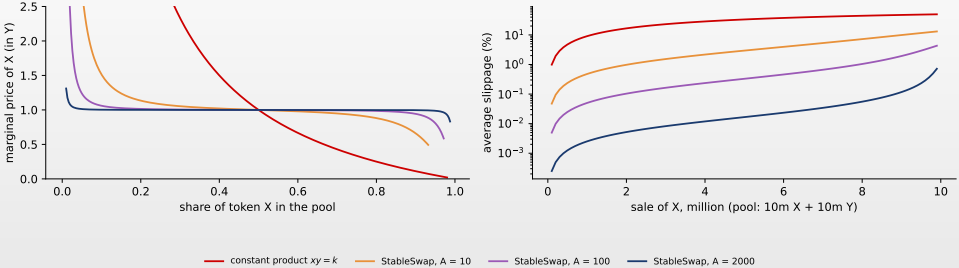
$$A\,n^n \sum_i x_i + D = A\,D\,n^n + \frac{D^{n+1}}{n^n \prod_i x_i}$$

  - $A \to 0$: constant product; $A \to \infty$: constant sum ($x + y = D$, price exactly 1)

Sale of X (pool 10 m + 10 m)	constant product	$A = 10$	$A = 100$	$A = 2000$
0.1 m	0.99%	0.048%	0.005%	0.000%
1 m	9.09%	0.48%	0.05%	0.003%
5 m	33.3%	2.95%	0.33%	0.017%
8 m	44.4%	7.22%	1.04%	0.055%
9.5 m	48.7%	11.6%	2.94%	0.232%

- Average slippage (own computation): StableSwap is thousands of times cheaper near balance, but slippage rises steeply when the pool empties of one coin

Prices Are Inelastic until They Are Not



- With $A = 100$ the price of X is 0.986 when X is 80% of the pool, 0.943 at 90%, 0.807 at 95%; with $A = 2000$: 0.999, 0.997, 0.988
- The pool absorbs one-sided flow with almost no price change, so the *price* hides the stress; the *composition* shows it

Data and Target in the Paper

- Chaffard, Găman, Staures, Ginavar, Jheng, Härdle & Pele (2026): Uniswap V3 USDC/USDT pool (0.01% fee), hourly, January 2022 – May 2026, 38 045 observations
- Feature blocks: pool price and flows; Curve 3pool shares and entropy; swap-impact curves; Gegenbauer scores; ownership (JIT share, Herfindahl–Hirschman index (HHI), width, age of positions); Aave utilisation, rates, liquidations; BTC/ETH volatility; a USD index from Chainlink; fear and greed index
- Target: a breach of $\delta = 15$ bp within the next 24 hours

$$y_t = \mathbb{1}\Big\{ \max_{1 \le h \le 24} |d_{t+h}| \ge \delta \Big\}$$

- Labels vs events
  - paper (pool): 2 227 positive hours (5.9%) but only 107 events
  - own (EODHD cross rate USDC/USDT, Jan 2022 – Oct 2026): 3 010 positive hours (7.1%), 50 episodes
  - the venue matters: the pool shows about twice as many events as the aggregated price (episode rules differ slightly)

Episodes by Threshold (own)

δ (bp)	episodes	breach hours	2022	2023	2024	2025	2026
10	119	4 282	37	10	38	14	20
15	50	1 044	21	7	14	3	5
25	8	358	4	2	1	1	0
50	3	105	2	1	0	0	0

- USDC priced in USDT, hourly EODHD prices; episode = first breach hour after 24 quiet hours
- Between 10 and 15 bp the number of episodes falls by more than half; above 25 bp there are only 8 in five years
- The threshold is a policy choice: low δ gives many, mostly harmless events; high δ leaves too few events to learn or to test

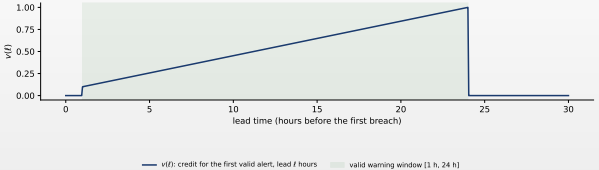
Evaluating an Alarm, Not a Classifier

- Row-level area under the receiver operating characteristic curve (AUC) and under the precision–recall curve (AUPRC) reward correct labels *during* an episode; an alert after the breach is no warning
- Event-level protocol
  - alerts: hours with score above a threshold; a 24-hour refractory period merges them into alert episodes
  - event e at time τ_e is detected if an alert falls in $[\tau_e - 24\text{h}, \tau_e - 1\text{h}]$; lead ℓ_e = hours before τ_e
  - all other alert episodes are false; F_m = false alerts per month
- Utility

$$U = \frac{1}{|E|} \sum_{e \in E} v(\ell_e) - 0.05\, F_m, \qquad v(\ell) = 0.10 + 0.90\, \frac{\ell - 1}{23} \ \text{ for } 1 \le \ell \le 24, \ \text{else } 0$$

- Alert threshold chosen on the preceding validation fold, maximising U subject to $F_m \le 2$ per month; never re-tuned on the test fold

Worked Example: Computing the Utility

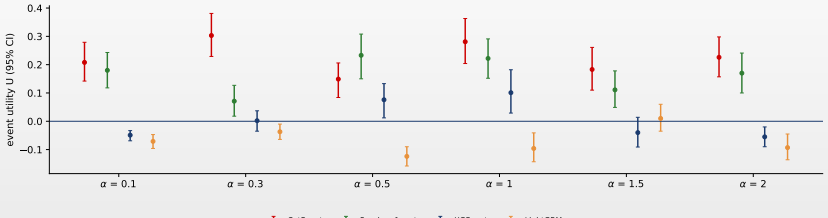


- Four events; first valid alerts 14 h, 2 h, none, 20 h before the breach; 2.5 false alerts per month

$$v = (0.609,\ 0.139,\ 0,\ 0.843), \qquad \bar{v} = 0.398, \qquad U = 0.398 - 0.05 \cdot 2.5 = 0.273$$

- One more false alert per month costs 0.05, about the value of detecting one extra event (out of four) with a 2-hour lead

Results: Four Classifier Families (Paper, Table 1)



	U [95% CI]	recall	F_m	median lead (h)	AUC	AUPRC
CatBoost, $\alpha = 0.3$	0.303 [0.229, 0.381]	0.723	2.50	14.4	0.862	0.241
Random forest, $\alpha = 0.1$	0.180 [0.118, 0.243]	0.410	0.97	12.0	0.901	0.239

- CatBoost detects 72% of events, 14 hours ahead; the random forest with the *best AUC* detects only 41%: AUC is not the right criterion

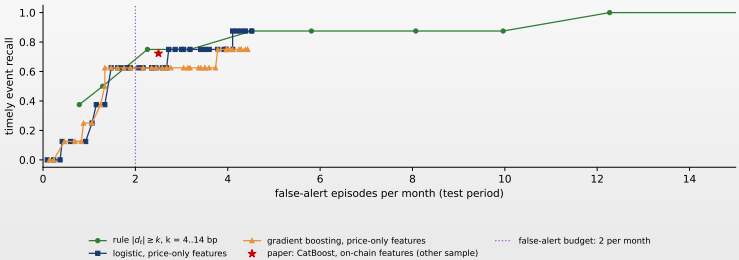
A Price-Only Baseline (own)

- Same protocol on the EODHD cross rate: train 2022–2023, validation 2024 (threshold, $F_m \leq 2$), test 2 Jan 2025 – 23 Oct 2026, 24-hour embargoes
- Features: d_t, $|d_t|$, 24 h max and volatility of d, deviation from the weekly mean, DAI and USDT vs USD, BTC 24 h return and volatility

Test, 8 events	detected	recall	lead (h)	F_m	U	AUC	AUPRC
Logistic regression	1	0.13	2	0.60	−0.013	0.928	0.308
Gradient boosting	0	0	–	0.00	0.000	0.898	0.325
Rule $\|d_t\| \geq k$, $k = 20$ chosen in 2024	0	0	–	0.19	−0.009	0.905	0.411

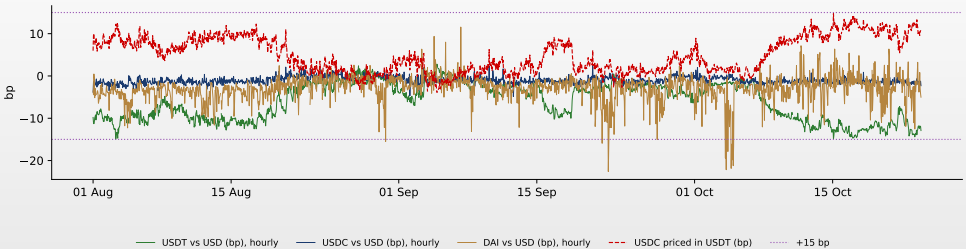
- AUC above 0.89 everywhere, yet almost no warnings: high AUC comes from labelling hours *inside* episodes
- In 2024 every rule that detected events produced at least 2.6 false alerts a month, so the budget forced the threshold above 15 bp

The Recall–False-Alert Trade-off (own)



- With hindsight, the rule $|d_t| \geq 11$ bp would have reached recall 0.75, lead 17.5 h, $U = 0.30$, but with 3.2 false alerts a month and k chosen on the test data
- 6 of 8 events: Clopper–Pearson 95% interval for recall $[0.35, 0.97]$. With so few events, the ranking of alarms is fragile

The Last Twelve Weeks (own)



- USDT below USD: −6.6 bp on average in August, −3.2 in September, −8.7 in October 2026
- USDC priced in USDT: October mean 7.5 bp, maximum 14.7 bp (15 October, 02:00 UTC); 206 of 552 hours above 10 bp, none above 15 bp
- A rule at 10 bp would have fired repeatedly; the paper's alarm would look at the pool composition and the liquidity curve before calling it stress

Summary

- Peg deviations have mean zero and power-law tails; the risk is in rare episodes: 50 breaches of 15 bp in USDC/USDT since 2022, 3 beyond 50 bp
- SVB: USDC fell to $-1\,036$ bp, below its zero-recovery value (-825 bp): access and time matter as much as reserves
- AMMs keep prices flat until a pool is nearly one-sided: pool composition reveals stress before the price
- The paper's alarm detects 72% of events 14 hours ahead with 2.5 false alerts a month; price-only alarms meet the budget only by detecting almost nothing

Interpretation Exercises

1. USDC has the smallest 1% quantile of the three coins (-7.6 bp) but the largest kurtosis (1 045). What do the two numbers say together about its risk?
2. A Curve pool with $A = 2000$ shows a price of 0.997 while 90% of it is USDC. Is USDC safe? What would you monitor instead of the price?
3. The random forest has AUC 0.901 and recall 0.41; CatBoost has AUC 0.862 and recall 0.72. Which would you deploy, and why does the ranking differ?
4. A rule $|d_t| \geq 11$ bp reaches $U = 0.30$ on the test set. Give two reasons why this does not show that the rule is as good as the paper's model.

⊡ Next: Lecture 5, the De-pegging Pressure Index across stablecoins

Appendix: Further Material

Optional material for self-study. It is **not examined**.

- A Threshold Autoregression
- Estimates: Hourly USDC/USDT, 2022–2026 (own)
- The Fitted Band
- Worked Example: How Fast Does a Deviation Close?
- Uniswap V3: Concentrated Liquidity
- Curve 3pool: Entropy as an Imbalance Index
- A Fixed Basis for a Moving Curve
- Worked Example: Decomposing a Stylised Curve (own)
- What Drives the Warnings? SHAP Attribution
- From an Alarm to a Distribution
- A Quantile Function with Pareto Tails
- What the Tail Parameters Do (own illustration)
- Scoring a Distribution: CRPS and twCRPS
- Worked Example: Two Forecasters, Three Outcomes (own)
- Results: Distributional Accuracy (Paper, Tables 2–3)
- Calibration and Ablation

A Threshold Autoregression

- If arbitrage costs c, small deviations are not worth trading; beyond c they are. A threshold autoregressive (AR) model captures this:

$$\Delta d_t = \mu + \rho_{in}\, d_{t-1}\, \mathbb{1}\{|d_{t-1}| \le \gamma\} + \rho_{out}\, d_{t-1}\, \mathbb{1}\{|d_{t-1}| > \gamma\} + \varepsilon_t$$

  - band hypothesis: $\rho_{in} \approx 0$, $\rho_{out} < 0$; half-life $h = \ln 0.5 / \ln(1+\rho)$
- Estimation: for each γ on a grid, ordinary least squares (OLS); $\hat{\gamma} = \arg\min_\gamma \mathrm{SSR}(\gamma)$, the sum of squared residuals (SSR)
- Test of a threshold effect, $H_0 : \rho_{in} = \rho_{out}$; γ is not identified under H_0, so the usual χ^2 fails; with a heteroskedasticity-consistent (HC) variance:

$$\sup W = \sup_{\gamma \in \Gamma} W(\gamma), \qquad W(\gamma) = \frac{(\hat{\rho}_{in} - \hat{\rho}_{out})^2}{\widehat{\mathrm{Var}}_{\mathrm{HC}}(\hat{\rho}_{in} - \hat{\rho}_{out})}$$

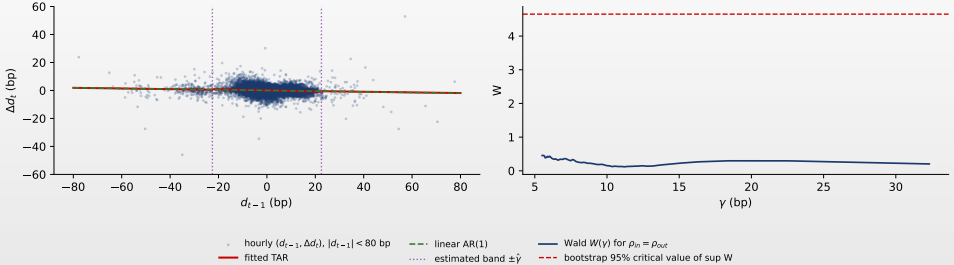
- p-value by the fixed-regressor bootstrap (Hansen (1996)): $y_t^* = \hat{e}_t z_t$, $z_t \sim N(0,1)$, recompute $\sup W$, $B = 199$

Estimates: Hourly USDC/USDT, 2022–2026 (own)

	coefficient	SE (HC)	half-life (hours)
$\hat{\gamma}$ (bp)	22.5		99.0% of hours inside
$\hat{\rho}_{in}$	−0.0396	0.0020	17.2
$\hat{\rho}_{out}$	−0.0233	0.0297	29.4
linear AR(1): $\hat{\rho}$	−0.0245		28.0
sup W (bootstrap 95% critical value)	0.5	(4.7), p = 0.653	

- $n = 40\,545$ consecutive hourly pairs; 2.8% of hours missing in the source; SE: standard error
- No threshold effect: we cannot reject $\rho_{in} = \rho_{out}$; deviations decay in about a day either way
- The band story fails in this sample because the large deviations are SVB and Terra hours, when redemption was closed or trust was gone: arbitrage was not available, so reversion was *not* faster
- Lesson: a band model describes normal times; stress is a change of regime, not a larger deviation of the same process

The Fitted Band



- Left: the hourly change against last hour's deviation; slopes are small (≈ -0.04): reversion is slow at the hourly scale
- Right: the Wald profile $W(\gamma)$ never comes close to the bootstrap critical value

Worked Example: How Fast Does a Deviation Close?

- Inside the band, $\hat{\rho}_{in} = -0.0396$: the expected deviation after h hours is

$$E[d_{t+h} \mid d_t] \approx (1 + \hat{\rho}_{in})^h \, d_t \quad (\hat{\mu} \approx 0)$$

- Start from $d_t = -12$ bp (USDT cheap against USDC, as on 23 October 2026):

$$\text{24 hours: } 0.9604^{24} = 0.379, \qquad E[d_{t+24}] \approx -4.5 \text{ bp}$$

- Half-life $\ln 0.5 / \ln 0.9604 = 17.2$ hours
- Interpretation
  - a deviation of this size is expected to shrink by about 60% in a day
  - if it does not, the process is not "inside the band": that persistence is itself a warning signal

Uniswap V3: Concentrated Liquidity

- Liquidity providers choose a price range; prices live on ticks $p_k = 1.0001^k$, one tick = 1 bp
- Active liquidity at tick j: $L_{t,j}$ = sum of all positions whose range contains j; the curve $j \mapsto L_{t,j}$ changes every hour
- Swap-impact curves from the current tick k_t (Chaffard, Găman, Staures, Ginavar, Jheng, Härdle & Pele (2026)):

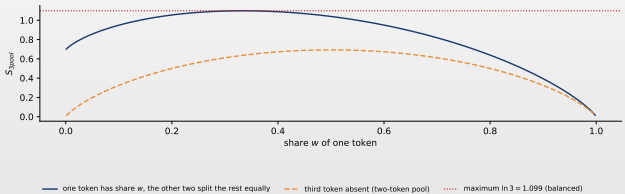
$$Q_t^+(k) = \sum_{j=k_t}^{k} L_{t,j} \;\; (k > k_t), \qquad Q_t^-(k) = \sum_{j=k}^{k_t} L_{t,j} \;\; (k < k_t)$$

  - converted to the token amount needed to move the price to tick k
  - area under $Q^{\pm}$: global depth on each side; slope at k_t: local depth
  - a sudden flat part: a **liquidity cliff**, where a small trade moves the price a lot
- Data: The Graph subgraphs, hourly, ± 50 ticks around the peg; positions minted and burned in the same block are just-in-time (JIT) liquidity and may not be there under stress

Curve 3pool: Entropy as an Imbalance Index

- Normalised balances $w_{\mathrm{DAI}} + w_{\mathrm{USDC}} + w_{\mathrm{USDT}} = 1$:

$$S_{3pool} = -\sum_i w_i \ln w_i, \qquad 0 \le S \le \ln 3 = 1.0986$$



- Worked example: balanced $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$: $S = 1.0986$; $(0.20, 0.40, 0.40)$: 1.0549; $(0.05, 0.10, 0.85)$: 0.5182
- Flat near balance, steep near a corner: small changes in S at high levels are noise, a fall below about 1 is stress

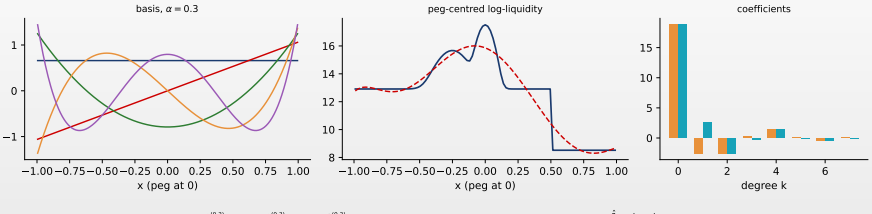
A Fixed Basis for a Moving Curve

- Peg-centred coordinate $x \in [-1, 1]$ (peg at $x = 0$), log-liquidity $\ell_t(x) = \log\{L_t(x) + 1\}$
- Expansion in Gegenbauer polynomials, orthonormal under the weight w_α:

$$\ell_t(x) \approx \sum_{k=0}^{K} \beta_{t,k}\, \phi_k^{(\alpha)}(x), \qquad w_\alpha(x) = (1 - x^2)^{\alpha - 1/2}, \qquad \beta_{t,k} = \int_{-1}^{1} \ell_t(x)\, \phi_k^{(\alpha)}(x)\, w_\alpha(x)\, dx$$

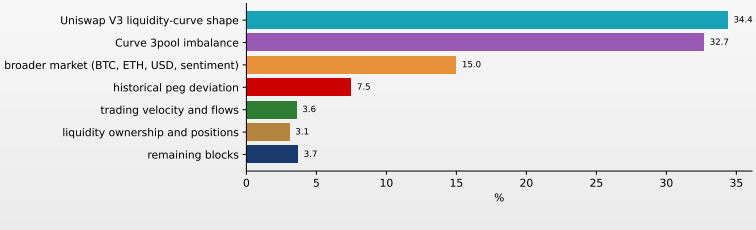
- Reading the coefficients
  - $k = 0$: level of liquidity; $k = 1$: left-right asymmetry; $k = 2$: concentration at the peg
  - higher k: local notches and cliffs; odd k asymmetric, even k symmetric
- Energies: $E_t^{odd} = \sum_{k \text{ odd}} \beta_{t,k}^2$, $E_t^{even} = \sum_{k \geq 2 \text{ even}} \beta_{t,k}^2$
- α moves the weight between centre and edges; it is tuned as a hyperparameter. A fixed basis keeps the scores comparable over time, unlike rolling functional principal component analysis (FPCA)

Worked Example: Decomposing a Stylised Curve (own)



- Stylised curve: a deep band at the peg, extra depth below parity, a cliff above $x = 0.5$; $\alpha = 0.3$, $K = 7$
- $\hat{\beta} = (18.78, -2.53, -2.50, 0.32, 1.45, 0.08, -0.43, 0.07)$; $E^{odd} = 6.48$, $E^{even} = 8.55$
- The mirror image has the *same* even coefficients and odd coefficients with the opposite sign: the odd scores carry the direction of fragility

What Drives the Warnings? SHAP Attribution



- SHapley Additive exPlanations (SHAP; Lundberg & Lee (2017)): $f(\mathrm{x}) = \phi_{\mathbf{0}} + \sum_j \phi_j(\mathrm{x})$, shares of mean $|\phi_j|$ grouped by economic block (interventional, main effects)
- Past peg deviations explain only 7.5%: the model warns from liquidity shape (34.4%) and 3pool imbalance (32.7%), not from the price itself
- Attribution, not causality: correlated on-chain features share credit

From an Alarm to a Distribution

- An alarm says *whether*; a risk manager also needs *how far*: the probability of crossing a liquidation level, the expected shortfall (ES) of a depeg
- The paper forecasts the full conditional distribution of d_{t+h}, $h = 1, \ldots, 24$
- Architecture, in brief
  - two branches: the target history (dilated causal convolutions) and the covariates (one token per variable)
  - reversible instance normalisation; a gated residual network scores covariates, sparsemax sets most weights to exactly zero
  - cross-attention from forecast horizons to the selected covariates; an output head returns a quantile function
- Benchmarks: naive, autoregressive integrated moving average (ARIMA), generalised autoregressive conditional heteroskedasticity with Student-t errors (GARCH-t), Time-series Dense Encoder (TiDE) and TimeXer; the two neural benchmarks get the same distributional head and loss
- Test period: January 2025 – May 2026

A Quantile Function with Pareto Tails

- Body: monotone I-spline, centred at the median and scaled

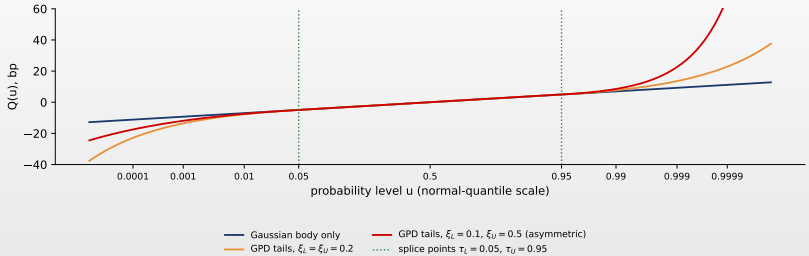
$$Q_t^0(u) = \sum_{k=1}^{K} a_{t,k} I_k(u),\ a_{t,k} > 0; \qquad Q_t^B(u) = \ell_t + \frac{s_t}{\sqrt{V_t}} \{Q_t^0(u) - Q_t^0(0.5)\}$$

- Tails: generalised Pareto distribution (GPD), joined smoothly at τ_L and τ_U

$$Q_t(u) = \begin{cases} z_L - \dfrac{\beta_L}{\xi_L}\left[\left(\frac{u}{\tau_L}\right)^{-\xi_L} - 1\right], & u < \tau_L \\ Q_t^B(u), & \tau_L \le u \le \tau_U \\ z_U + \dfrac{\beta_U}{\xi_U}\left[\left(\frac{1-u}{1-\tau_U}\right)^{-\xi_U} - 1\right], & u > \tau_U \end{cases}$$

- $z_L = Q_t^B(\tau_L)$, $\beta_L = \tau_L\, Q_t^{B\prime}(\tau_L)$ and symmetrically for the upper tail; separate ξ_L, ξ_U allow USDC and USDT depegs to differ
- For $\xi \ge 1/2$ the tail has infinite variance (Lecture 2: tail index $1/\xi$)

What the Tail Parameters Do (own illustration)



- Same Gaussian body (sd 3 bp), splice at 5% and 95%; on the normal-quantile scale the Gaussian is a straight line
- 99.9% quantile: Gaussian 9.3 bp; GPD with $\xi = 0.2$: 13.6 bp; with $\xi_U = 0.5$: 22.6 bp, while the lower tail ($\xi_L = 0.1$) stays at -11.9 bp

Scoring a Distribution: CRPS and twCRPS

- Continuous ranked probability score (Gneiting & Raftery (2007)), a proper scoring rule:

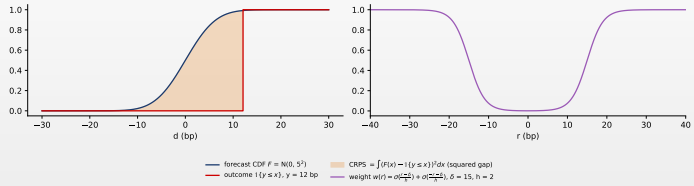
$$\mathrm{CRPS}(F, y) = \int_{-\infty}^{\infty} \big(F(x) - \mathbb{1}\{y \le x\}\big)^2 dx = 2\int_0^1 \rho_u\big(y - Q(u)\big)\, du$$

  - $\rho_u(e) = e\,(u - \mathbb{1}\{e < 0\})$: the pinball loss; CRPS averages it over all quantiles
  - Normal: $\mathrm{CRPS}(N(\mu, \sigma^2), y) = \sigma\big[z(2\Phi(z) - 1) + 2\varphi(z) - 1/\sqrt{\pi}\big]$, $z = (y - \mu)/\sigma$
- Threshold-weighted CRPS (Gneiting & Ranjan (2011)): put the weight where depegs are

$$\mathrm{twCRPS} = 2\int_0^1 \rho_u\big(v(y) - v(Q(u))\big)\, du, \quad v(r) = \int_0^r w(s)\, ds, \quad w(r) = \sigma\Big(\frac{r - \delta}{h}\Big) + \sigma\Big(\frac{-r - \delta}{h}\Big)$$

- The paper uses twCRPS as training loss and both scores for evaluation

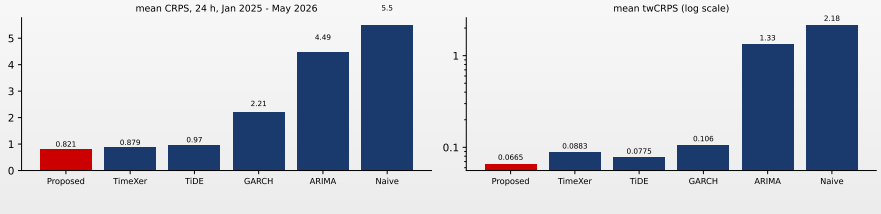
Worked Example: Two Forecasters, Three Outcomes (own)



forecast	CRPS $y=1$	CRPS $y=12$	CRPS $y=25$	twCRPS ($\delta=15$, $h=2$, illustrative) $y=1$	twCRPS ($\delta=15$, $h=2$, illustrative) $y=12$	twCRPS ($\delta=15$, $h=2$, illustrative) $y=25$
$N(0, 5^2)$	1.25	9.21	22.18	0.003	0.378	9.98
$N(0, 10^2)$	2.38	7.48	19.40	0.054	0.324	9.34
t_3, scale 3	0.95	9.71	22.57	0.003	0.381	9.91

- Calm hour ($y = 1$): the sharp forecasts win; depeg hour ($y = 25$): the wide one wins. twCRPS ignores calm hours (0.003) and scores only the depeg region

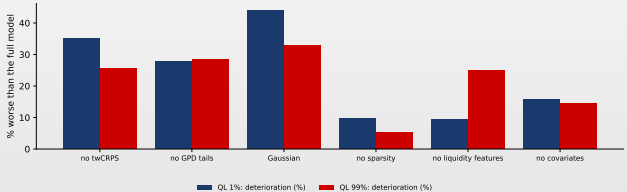
Results: Distributional Accuracy (Paper, Tables 2–3)



- Lowest loss on every metric; against the best benchmark: CRPS −6.6% (TimeXer), twCRPS −14.2% (TiDE)
- Against GARCH-t, from Table 2: CRPS 0.82 vs 2.21 (−63%), twCRPS 0.067 vs 0.106 (−37%)
- One-sided Diebold–Mariano tests with block-bootstrap intervals: all p-values below 0.05 against all five benchmarks

Calibration and Ablation

- Probability integral transform (PIT) $u_t = \hat{F}_t(y_t)$: mean 0.5022 (target 0.5), variance 0.0878 (target $1/12 = 0.0833$); Kolmogorov–Smirnov (KS) $D = 0.0288$, 48-hour block bootstrap p $= 0.079$
- Value-at-risk (VaR) exceedances: too many at the 2–5% levels in both tails, close to nominal at 0.5–1%; forecast ES more extreme than realised: conservative



- A Gaussian head loses 44% at the 1% quantile; dropping the liquidity features costs 25% at the 99% quantile, more than dropping all covariates